

spent time as an intellectual refugee at an IBM research center in upstate New York. It was a *f*** you money* situation, as IBM let him do whatever he felt like doing.

But the general public (mostly computer geeks) got the point. Mandelbrot's book *The Fractal Geometry of Nature* made a splash when it came out a quarter century ago. It spread through artistic circles and led to studies in aesthetics, architectural design, even large industrial applications. Benoît M. was even offered a position as a professor of medicine! Supposedly the lungs are self-similar. His talks were invaded by all sorts of artists, earning him the nickname the Rock Star of Mathematics. The computer age helped him become one of the most influential mathematicians in history, in terms of the applications of his work, way before his acceptance by the ivory tower. We will see that, in addition to its universality, his work offers an unusual attribute: it is remarkably easy to understand.

A few words on his biography. Mandelbrot came to France from Warsaw in 1936, at the age of twelve. Owing to the vicissitudes of a clandestine life during Nazi-occupied France, he was spared some of the conventional Gallic education with its uninspiring algebraic drills, becoming largely self-taught. He was later deeply influenced by his uncle Szolem, a prominent member of the French mathematical establishment and holder of a chair at the Collège de France. Benoît M. later settled in the United States, working most of his life as an industrial scientist, with a few transitory and varied academic appointments.

The computer played two roles in the new science Mandelbrot helped conceive. First, fractal objects, as we have seen, can be generated with a simple rule applied to itself, which makes them ideal for the automatic activity of a computer (or Mother Nature). Second, in the generation of visual intuitions lies a dialectic between the mathematician and the objects generated.

Now let us see how this takes us to randomness. In fact, it is with probability that Mandelbrot started his career.

A Visual Approach to Extremistan/Mediocristan

I am looking at the rug in my study. If I examine it with a microscope, I will see a very rugged terrain. If I look at it with a magnifying glass, the terrain will be smoother but still highly uneven. But when I look at it from a standing position, it appears uniform—it is almost as smooth as a sheet of paper. The rug at eye level corresponds to Mediocristan and the law of large numbers: I am seeing the sum of undulations, and *these iron out*. This is like Gaussian